



2025

N24

BOOKLET NO.

106630

Forest Services
Botany

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining questions, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

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P.T.O.



SECTION – A

Q1. Write short notes on **any five** from seven. **(8×5=40)**

Q1. (a) What are plastids ? Give an account of structure, function and chemical composition of chloroplasts. **8**

Q1. (b) Describe fungal toxins with reference to producer and diseases caused by them. **8**

Q1. (c) What is heterospory ? Comment on causes and significance of heterospory in Pteridophytes. **8**

Q1. (d) Draw and describe structure of ovule in Gnetum. **8**

Q1. (e) Suggest an important measures to control global warming. **8**

Q1. (f) Define meiosis. Enlist any seven differences between mitosis and meiosis. **8**

Q1. (g) Write critically on an endangered plants in India with examples. **8**

Q2. (a) Give an account of structural variations in chromosomes and their significance. **15**

Q2. (b) Draw and describe structure of generalised bacterial cell. Add a note on classification of bacteria based on shapes. **15**

Q2. (c) Give an account of economic importance of algae. **10**

Q3. (a) Give an account of evolution of sporophytes in Bryophytes. **15**

Q3. (b) What is Polyembryony ? State the classification and significance of Polyembryony. **15**

Q3. (c) What is Phytoremediation ? Give an account of different types of Phytoremediation. **10**

Q4. (a) Define ecosystem. Explain the structure of forest ecosystem in detail. **15**

Q4. (b) What are giant chromosomes ? Draw and describe structure of lampbrush chromosomes. **15**

Q4. (c) Describe any three fungal plant diseases with reference to causal organism, symptoms and its control measures. **10**

**SECTION – B**

- Q5.** Write short notes on **any five** from seven. **(8×5=40)**
- Q5.** (a) What is seed dormancy ? Discuss its causes. **8**
- Q5.** (b) Give an account of sex chromosomes. **8**
- Q5.** (c) What do you mean by Standard Deviation ? Mention the steps involved in calculation of Standard Deviation. **8**
- Q5.** (d) Describe the key differences between Somatic hybrids and cybrids. **8**
- Q5.** (e) Discuss the importance of Botanical gardens. **8**
- Q5.** (f) Explain the significance of linkage in genetics. **8**
- Q5.** (g) What is Heterosis ? Discuss its causes and applications in plant breeding. **8**
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- Q6.** (a) Explain the significance of respiration. Briefly analyze the factors affecting rate of respiration. **15**
- Q6.** (b) What do you understand by organic evolution ? Name some theories proposed to explain the process. Discuss the evidences for organic evolution. **15**
- Q6.** (c) Mention the botanical sources, plant parts used and economic importance of the following :
- i) Lemon grass
 - ii) Black pepper (Kali miri)
 - iii) Licquorice (Jeshthimadh)
 - iv) Cotton (Kapas)
 - v) Shisham (Sissoo)
- 10**



- Q7.** (a) Give a brief account of the important methods of plant breeding. **15**
- Q7.** (b) What are organ cultures ? Briefly discuss the different organ culture techniques. **15**
- Q7.** (c) Discuss cytoplasmic inheritance with the help of suitable examples. **10**
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- Q8.** (a) What are enzymes ? How are they classified ? **15**
- Q8.** (b) Discuss the genetic engineering methods used for transgenic crop development. Mention the steps involved in development of transgenic crops. **15**
- Q8.** (c) Explain the methodology and importance of plant protoplast cultures. **10**
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